

Brijodison

Engineering Notebook Element B

Engineering Service Learning

Existing Solutions

There appears to be a few solutions to the problem, but there doesn't appear to be very much information on the development of such solutions. A search on the U.S. patent database doesn't appear to reveal any solutions related to the problem, but I did find a few unpatented solutions and quite a few copies of the same unpatented solution. Not many people appear to have designed a specific solution to materials carts, but there are a plethora of solutions to the general problem.



Fig 1. Amazon

The first product offered by TOPofly shows a potential solution for a suitcase that inspired our own solution. It appears to be adapted from straight bike handles, with a straight pole extending down to extend the length. At the end, there is a clamp that inspired our solution

greatly. The clamp appears to be relatively short because a suitcase doesn't have an extended section to which the clamp can connect. The bar appears to be more flexible around the clamp section to adapt to a very slightly different handle. It's made of a rigid material to prevent the handle from wobbling. We took a few ideas from this solution, opting to add a clamp that followed a very similar shape and designing around a rigid metal that would withstand the force applied without bending.

Stroller Handle Extender



The Stroller Handle Extender is made of light-weight steel with an extra-wide foam grip to make pushing comfortable. This extension handle attaches quickly and securely to the stroller's handle with 4 industrial-strength Velcro® straps, adding approximately 8" of height and depth to the handle so you can stop kicking the stroller as you walk. [Check out our how-to guide](#) to help you easily connect the extender to your stroller.

Lawn Mower Handle Extender



Does mowing your 'corner of heaven' leave you with back and neck strain? Most push-type lawn mowers are made for people shorter than 5'8". Taller people lean forward and down because the mower handle is too low for them. Then they hunch their shoulders and "put some muscle" into cutting the grass. This unnatural posture, combined with the pressure required to push the mower over uneven terrain, results in fatigue, neck, shoulder and back strain.

The Lawn Mower Handle Extender attaches quickly and securely to your mower's handle with sturdy Velcro® straps. It lengthens the handle by approximately 8" allowing taller people like you to walk comfortably upright.

The Lawn Mower Handle Extender is made of light-weight steel with an extra-wide foam grip to make pushing comfortable. This extension handle attaches quickly and securely to the mower's handle with 4 industrial-strength Velcro® straps.

Note: If your mower comes equipped with an auto-shut-off bar: The Lawn Mower Handle Extender comes with a cord (and instructions) to extend the auto-shut-off safety feature too!

Fig 2. Handle Extender

We found an aptly named, “Handle Extender” online for strollers and lawnmowers. They opted to design their solutions around light-weight steel, which confirmed our decision to use the same. We noticed they used velcro, but we decided not to because it would require Mr. Kiser to secure the handle at a lower point, which would fail our criteria. Nonetheless, we noticed that this is a solution and considered ways to remotely attach a similar clamp. We particularly took a look at the handles which appeared to be designed for comfort and noticed that the common theme was compressibility. This would eventually lead to the final solution we would use for the handle. (We used shirts to mimic the same compressibility.)



Fig 3. Haul Pro

We found a few more specialized solutions for particular carts and decided to take a look nonetheless. We knew that we would have to come up with a different design for the clamp, but we still figured we could gain something by observing other solutions. This particular solution slides in and out of a specific cart, allowing the user to adjust it to set heights. This solution uses metal to ensure rigidity and follows with a horizontal bar upon which the user may rest their

hands. Its simplicity speaks wonders and somewhat encouraged us to design our handle very simply as well.



Question: Are the clamps metal or plastic?

Answer: The clamps are made of tempered steel and then coated in a non-slip vinyl. They are designed to firmly grip onto your wheelchair without damaging it.

Fig 4. Travel Wheelchair

We found that extending handlebars applied not only to carts but also to wheelchairs. We found a potential solution online that helped in the design of our handles greatly. When we eventually had a few problems with the stability of the solution, we thought back to our knowledge of this product, thinking that the non-slip vinyl might come in handy. This applies particularly well to the current situation because we know that we can't make any permanent changes to the cart. We don't particularly like the direction of the handlebars, however, because they appear to be intended to allow the person steering the wheelchair to lean over with their weight on the wheelchair. This is counterintuitive in our case because we don't want to encourage the user to bend over and potentially hurt their back more.



Fig 5. WONDERFOLD

We found another solution for a very particular type of stroller that appeared very comfortable. The handle has a covering that goes over it, which inspired our own solution. The handle also has a slight curve towards the end, which is interesting because the handle is tubular, so rotating it does absolutely nothing. Perhaps this is designed to lower the angle the pushing force of the person acts with respect to the horizontal. We observed this element and tried designing our handle to do the same, making the handles roughly vertical so the maximum component of the force acts to push the cart.



Fig 6. Strictly USA

We found one final design that inspired our own. We didn't particularly like the clamp design because it didn't seem to attach to the posts of the suitcase, and we couldn't apply this clamp solution anyways to the cart because the posts are too far apart. We did, however, like the handle that they used for this solution. We found a few other carts that used a vertical handle, and this handle applied that solution greatly, inspiring us to do the same with our own design.

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